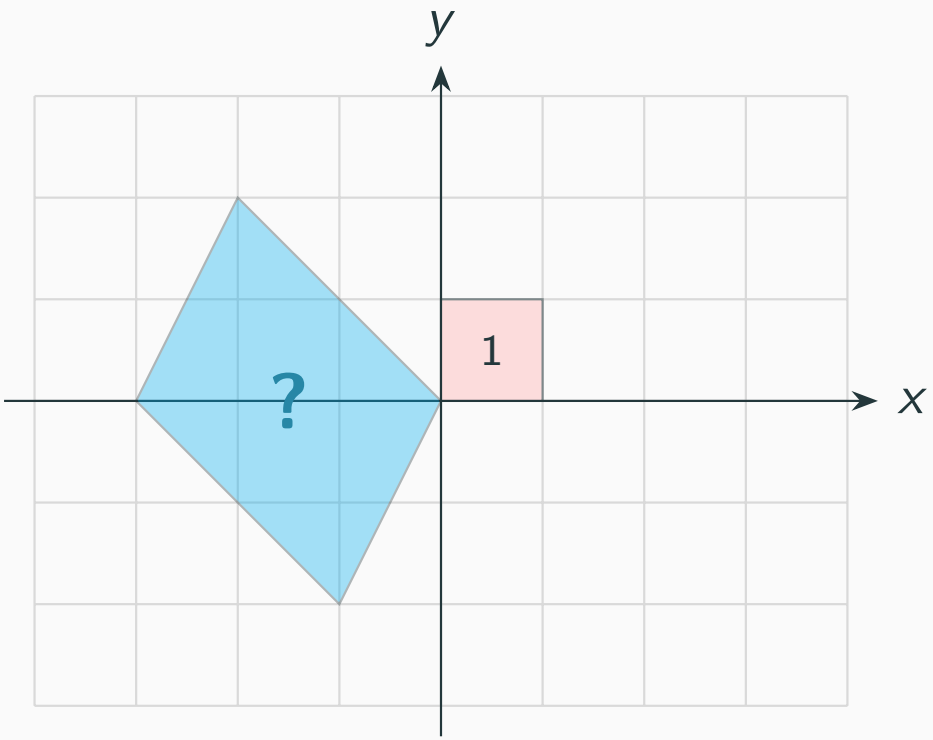
Lecture 12: Determinant

MATH203 Applied Linear Algebra

From Area to Determinant

How Big Is This Shape?

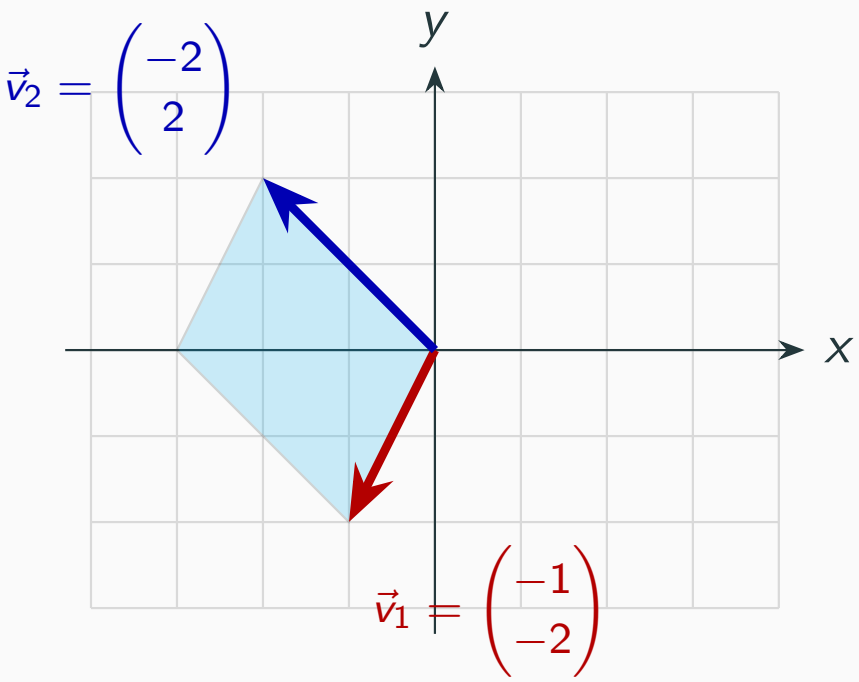
Suppose each small square on the grid has area 1.



How do you calculate the **blue area**?

Two Vectors Determine the Shape

The blue shape is a **parallelogram** — determined by two side vectors:

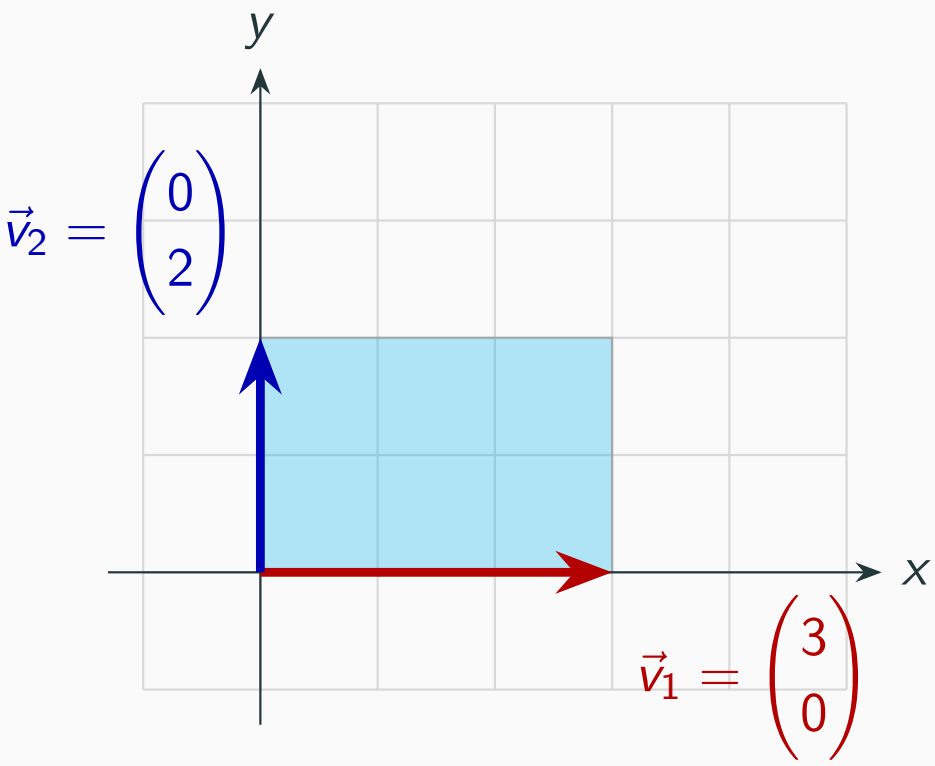


We write: $\det \begin{pmatrix} -1 & -2 \\ -2 & 2 \end{pmatrix} =$ **signed blue area**

"det" eats two column vectors and returns the **signed area** of their parallelogram.

The Easy Case: Axis-Aligned Rectangle

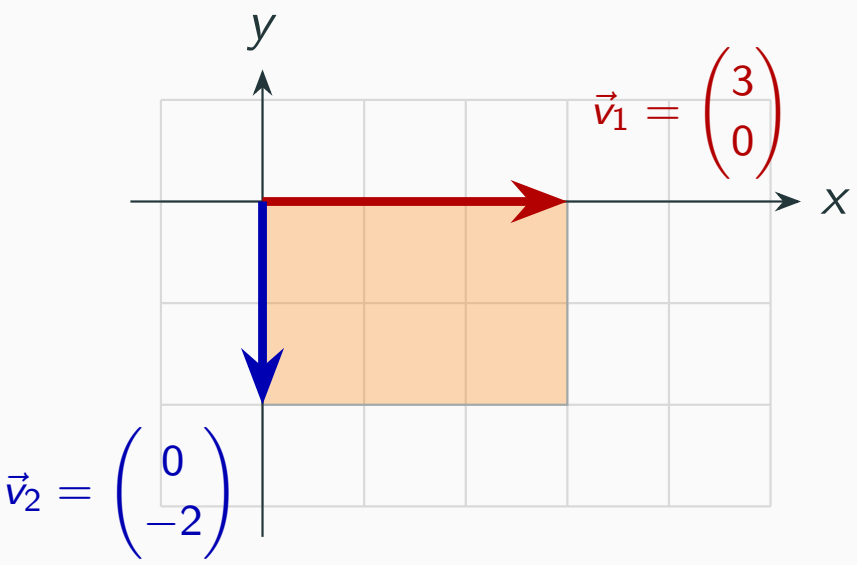
When both vectors point along the axes, the parallelogram is a **rectangle**:



$$\text{Area} = \text{base} \times \text{height} = 3 \times 2 = \mathbf{6}$$

What If the Height Points Down?

Now $\vec{v}_2$ points **downward**:

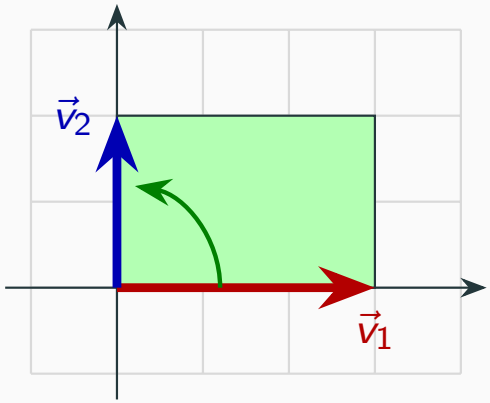


$$\text{Signed area} = \text{base} \times \text{height} = 3 \times (-2) = \textbf{-6}$$

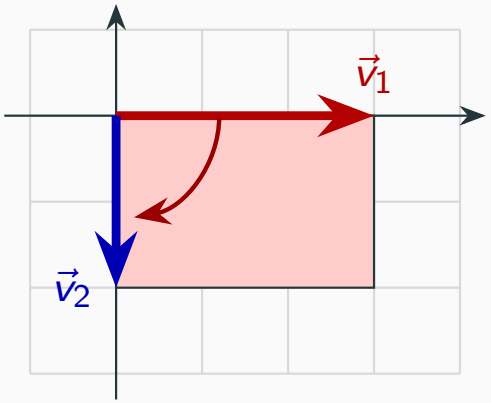
Same size, but the sign tells us the **orientation** has changed.

Oriented Area

This signed area is called the **oriented area**.



$\vec{v}_2$ is **counterclockwise** from $\vec{v}_1$: Area > 0

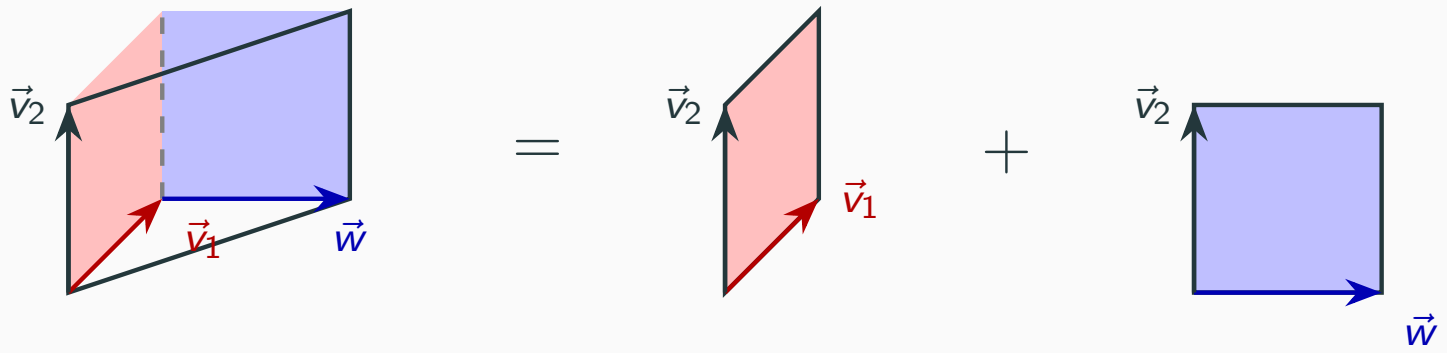


$\vec{v}_2$ is **clockwise** from $\vec{v}_1$: Area < 0

What **properties** should "signed area" satisfy?

Property 1: Additivity

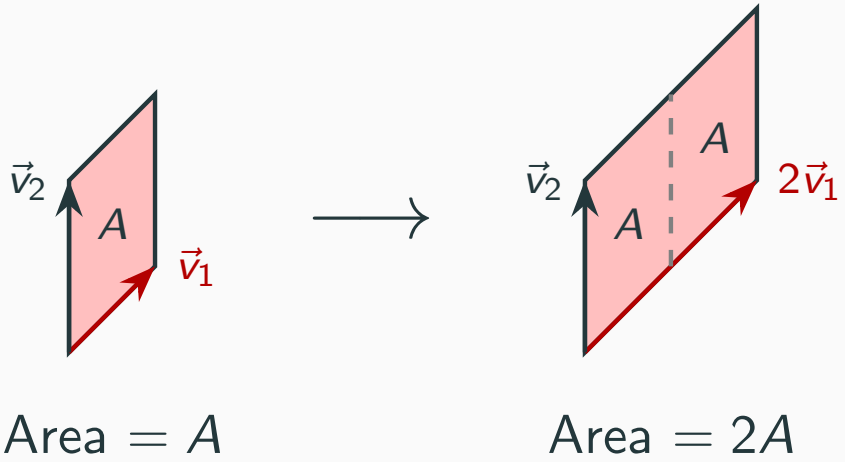
Fix the base $\vec{v}_2$. If we **add** the heights, the areas **add**:



$$\det\begin{pmatrix}\vec{v}_1 + \vec{w} & \vec{v}_2\end{pmatrix} = \det\begin{pmatrix}\vec{v}_1 & \vec{v}_2\end{pmatrix} + \det\begin{pmatrix}\vec{w} & \vec{v}_2\end{pmatrix}$$

Property 1: Scaling

If you **stretch** one side by factor k, the area scales by k:

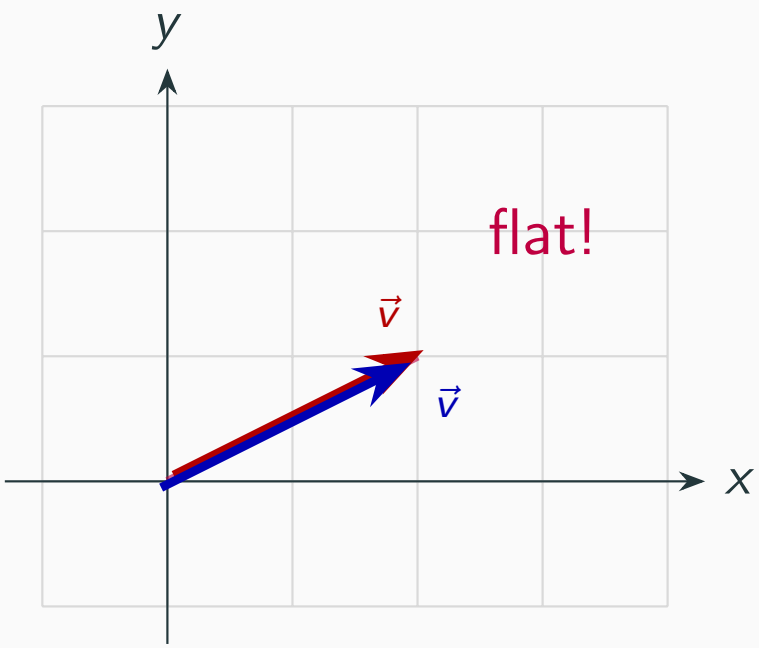


$$\det\begin{pmatrix} k\vec{v}_1 & \vec{v}_2 \end{pmatrix} = k \cdot \det\begin{pmatrix} \vec{v}_1 & \vec{v}_2 \end{pmatrix}$$

Together, additivity + scaling = **multilinearity**.

Property 2: Same Columns $\implies$ Zero Area

If both sides point in the **same direction**, the parallelogram collapses to a **line**:



$$\det\begin{pmatrix}\vec{v} & \vec{v}\end{pmatrix} = 0$$

A "parallelogram" with two identical sides has **zero area**.

Consequence: Swapping Columns Flips the Sign

Start from Property 2: $\det\begin{pmatrix}\vec{v}_1+\vec{v}_2 & \vec{v}_1+\vec{v}_2\end{pmatrix} = 0$

Expand by **multilinearity** (Property 1) in both columns:

$$\underbrace{\det\begin{pmatrix}\vec{v}_1 & \vec{v}_1\end{pmatrix}}_{=0} + \det\begin{pmatrix}\vec{v}_1 & \vec{v}_2\end{pmatrix} + \det\begin{pmatrix}\vec{v}_2 & \vec{v}_1\end{pmatrix} + \underbrace{\det\begin{pmatrix}\vec{v}_2 & \vec{v}_2\end{pmatrix}}_{=0} = 0$$

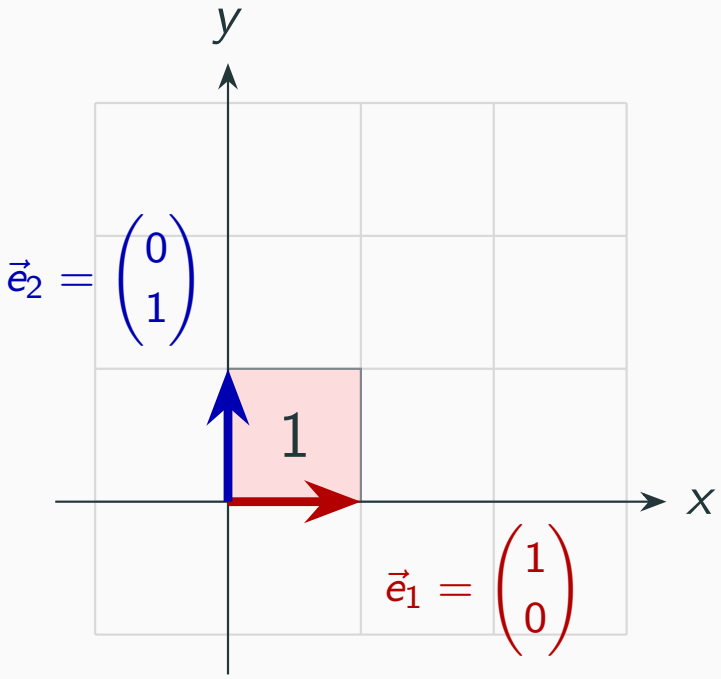
Therefore:

$$\boxed{\det\begin{pmatrix}\vec{v}_1 & \vec{v}_2\end{pmatrix} = -\det\begin{pmatrix}\vec{v}_2 & \vec{v}_1\end{pmatrix}}$$

Swapping the two columns **flips** the sign of the determinant.

Property 3: The Unit Square Has Area 1

The unit square spanned by $\vec{e}_1$ and $\vec{e}_2$ has area 1:



$$\det \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \det(I_2) = 1$$

Definition of Determinant

Definition

A function $\det\colon \{\text{square matrices}\} \to \mathbb{R}$ is called a **determinant** if it satisfies:

P1. Multilinearity: det is linear in each column separately:

$$\det(\ldots,\ k\vec{v} + \vec{w},\ \ldots) = k\det(\ldots,\ \vec{v},\ \ldots) + \det(\ldots,\ \vec{w},\ \ldots)$$

P2. Alternating: If two columns are equal, $\det = 0$.

P3. Normalization: $\det(I_n) = 1$.

These three properties **axiomatize** the notion of signed volume.

The 2×2 Case: Can We Find a Formula?

For a 2×2 matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$, what is det?

Two independent roads to the answer:

Road 1: Geometric

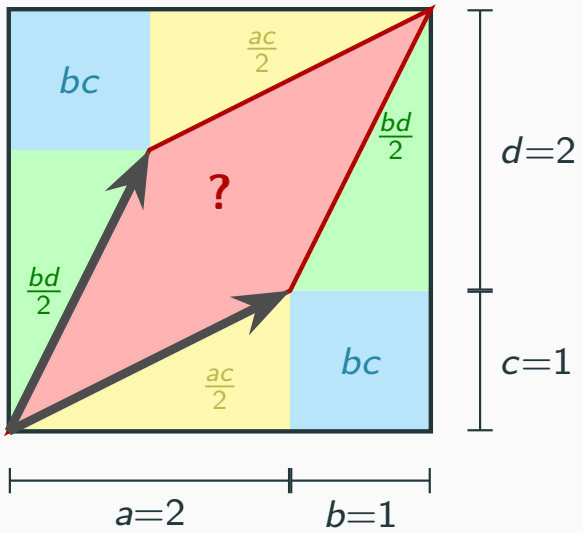
Enclose the parallelogram in a rectangle and subtract the remaining pieces

Road 2: Algebraic

Expand using P1, P2, P3 to derive the formula from the axioms directly

Both roads must give the **same answer** — because the formula is **uniquely** determined by P1–P3.

Road 1: The Rectangle Cut-Out



$$\underbrace{(a+b)(c+d)}_{\text{rectangle}} - ac - bd - 2bc = ?$$

Road 1: The Formula from Geometry

Expand the rectangle cut-out:

$$\begin{aligned} & (a+b)(c+d) - ac - bd - 2bc \\ = \; & ac + ad + bc + bd - ac - bd - 2bc \\ = \; & ad - bc \end{aligned}$$

Proposition

$$\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc$$

Road 2: Deriving from the Axioms

Write each column in terms of $\vec{e}_1, \vec{e}_2$:

$$\vec{v}_1 = \begin{pmatrix} a \\ c \end{pmatrix} = a\vec{e}_1 + c\vec{e}_2, \qquad \vec{v}_2 = \begin{pmatrix} b \\ d \end{pmatrix} = b\vec{e}_1 + d\vec{e}_2$$

Expand by **multilinearity** (P1) in the **first** column:

$$\det\begin{pmatrix} \vec{v}_1 & \vec{v}_2 \end{pmatrix} = \det\begin{pmatrix} a\vec{e}_1 + c\vec{e}_2 & \vec{v}_2 \end{pmatrix}$$

$$= a \cdot \det\begin{pmatrix} \vec{e}_1 & \vec{v}_2 \end{pmatrix} + c \cdot \det\begin{pmatrix} \vec{e}_2 & \vec{v}_2 \end{pmatrix}$$

Road 2: Applying P2 and P3

Now expand the **second** column in each term:

$$= ab \underbrace{\det\begin{pmatrix}\vec{e}_1 & \vec{e}_1\end{pmatrix}}_{= 0 \text{ (P2)}} + ad \underbrace{\det\begin{pmatrix}\vec{e}_1 & \vec{e}_2\end{pmatrix}}_{= 1 \text{ (P3)}}$$

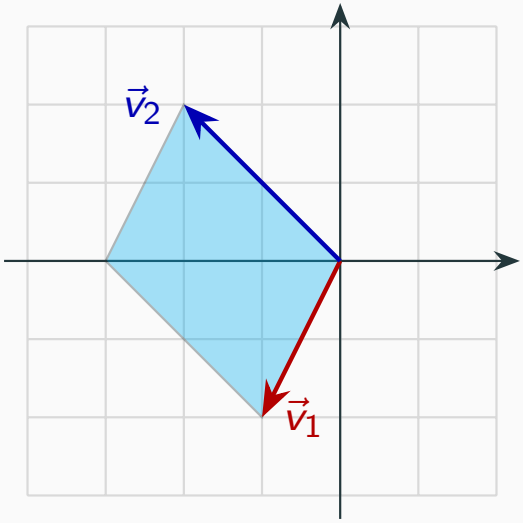
$$+ cb \underbrace{\det\begin{pmatrix}\vec{e}_2 & \vec{e}_1\end{pmatrix}}_{= -1 \text{ (swap)}} + cd \underbrace{\det\begin{pmatrix}\vec{e}_2 & \vec{e}_2\end{pmatrix}}_{= 0 \text{ (P2)}}$$

Therefore:

$$\det\begin{pmatrix}a & b \\ c & d\end{pmatrix} = 0 + ad \cdot 1 + cb \cdot (-1) + 0 = ad - bc \qquad \checkmark$$

Both roads arrive at the **same formula** — as they must.

Solving the Warm-Up



$$\det\begin{pmatrix} -1 & -2 \\ -2 & 2 \end{pmatrix}$$

$$= (-1)(2) - (-2)(-2)$$

$$= -2 - 4 = \textbf{-6}$$

$$\text{Area} = |-6| = \textbf{6}$$

The negative sign: $\vec{v}_2$ is **clockwise** from $\vec{v}_1$.

Practice

Using the formula $\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc$, calculate:

1. $\det \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix} =$
2. $\det \begin{pmatrix} 2 & 1 \\ 3 & 2 \end{pmatrix} =$
3. $\det \begin{pmatrix} 1 & 0 \\ 8 & 0 \end{pmatrix} =$
4. What does it mean geometrically when $\det = 0$?

Common Mistakes

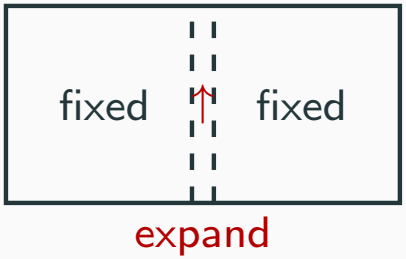
Multilinearity is Per-Column

Recall P1 (multilinearity):

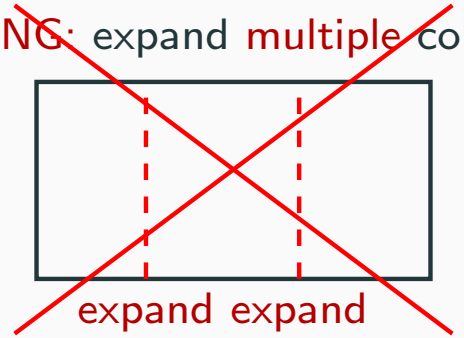
$$\det(\ldots,\ k\vec{v}+\vec{w},\ \ldots) = k\det(\ldots,\ \vec{v},\ \ldots) + \det(\ldots,\ \vec{w},\ \ldots)$$

The key word: **one column at a time**, other columns **stay fixed**.

Correct: expand one column



WRONG: expand multiple columns



Spot the Error

WRONG!

$$\det\begin{pmatrix} 1+2 & 1+3 \\ 2+1 & 3+2 \end{pmatrix} \stackrel{?}{=} \det\begin{pmatrix} 1 & 1 \\ 2 & 3 \end{pmatrix} + \det\begin{pmatrix} 2 & 3 \\ 1 & 2 \end{pmatrix}$$

Spot the Error

WRONG!

$$\det\begin{pmatrix}1+2 & 1+3\\ 2+1 & 3+2\end{pmatrix} \stackrel{?}{=} \det\begin{pmatrix}1 & 1\\ 2 & 3\end{pmatrix} + \det\begin{pmatrix}2 & 3\\ 1 & 2\end{pmatrix}$$

Check: $\text{LHS} = \det\begin{pmatrix}3 & 4\\ 3 & 5\end{pmatrix} = 15 - 12 = 3$

$\text{RHS} = (3-2) + (4-3) = 1 + 1 = 2 \neq 3$

The error: expanded **both columns simultaneously**.

Why It Fails: The Factor Analogy

Think of ordinary multiplication:

$$(x + y)(a + b) = xa + xb + ya + yb$$

You **cannot** skip the cross terms:

$$(x + y)(a + b) \neq xa + yb$$

Same with det: expanding **two** columns at once produces **four** terms, not two:

$$\det\begin{pmatrix} \vec{v}_1 + \vec{w}_1 & \vec{v}_2 + \vec{w}_2 \end{pmatrix}$$
$$= \det\begin{pmatrix} \vec{v}_1 & \vec{v}_2 \end{pmatrix} + \det\begin{pmatrix} \vec{v}_1 & \vec{w}_2 \end{pmatrix} + \det\begin{pmatrix} \vec{w}_1 & \vec{v}_2 \end{pmatrix} + \det\begin{pmatrix} \vec{w}_1 & \vec{w}_2 \end{pmatrix}$$

Another Trap: $\det(kA) \neq k\det(A)$

WRONG!

$$\det(2A) \stackrel{?}{=} 2\det(A)$$

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WRONG!

$$\det(2A) \stackrel{?}{=} 2\det(A)$$

If A is 3×3 with columns $\vec{v}_1, \vec{v}_2, \vec{v}_3$, then $2A$ has columns $2\vec{v}_1, 2\vec{v}_2, 2\vec{v}_3$.

$$\det(2A) = \det(2\vec{v}_1,\ 2\vec{v}_2,\ 2\vec{v}_3)$$

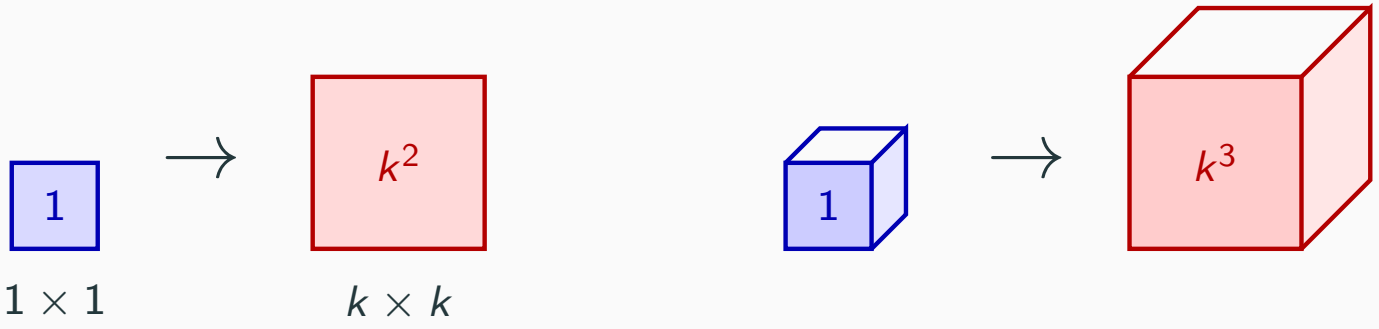
$$= 2\det(\vec{v}_1,\ 2\vec{v}_2,\ 2\vec{v}_3) = 4\det(\vec{v}_1,\ \vec{v}_2,\ 2\vec{v}_3) = \mathbf{8}\det(A)$$

Proposition

For an $n \times n$ matrix: $\det(kA) = \mathbf{k^n}\det(A)$

Why k^n? Think Geometrically

Scaling every column by k stretches **each dimension** by k.



- $n = 2$: area scales by k^2
- $n = 3$: volume scales by k^3
- n: "n-volume" scales by k^n

Practice: True or False?

Let A be a 4×4 matrix with $\det(A) = 3$.

1. $\det(2A) = 2 \cdot 3 = 6$?
2. $\det(2A) = 2^4 \cdot 3 = 48$?
3. $\det(-A) = -3$?
4. $\det(-A) = (-1)^4 \cdot 3 = 3$?

Column Operations and Switching Matrix

Column Operations Preserve Determinant

Theorem

Adding a multiple of one column to another does **not** change det:

$$\det(\ldots,\ \vec{v}_i + \lambda\vec{v}_j,\ \ldots,\ \vec{v}_j,\ \ldots) = \det(\ldots,\ \vec{v}_i,\ \ldots,\ \vec{v}_j,\ \ldots)$$

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Why? Expand by P1 (multilinearity in column i):

$$\det(\ldots,\ \vec{v}_i + \lambda\vec{v}_j,\ \ldots) = \det(\ldots,\ \vec{v}_i,\ \ldots) + \underbrace{\lambda \det(\ldots,\ \vec{v}_j,\ \ldots,\ \vec{v}_j,\ \ldots)}_{= 0 \text{ by P2 (two equal columns)}}$$

Example: Column Operations

$$\det\begin{pmatrix}1 & 3\\ 2 & 5\end{pmatrix} \overset{c_2-3c_1}{=} \det\begin{pmatrix}1 & 0\\ 2 & -1\end{pmatrix} = -1$$

More generally, for **any** number of columns:

$$\det\begin{pmatrix} | & | & & | \\ \vec{v}_1 & \vec{v}_2 & \cdots & \vec{v}_n \\ | & | & & | \end{pmatrix}$$

$$= \det\begin{pmatrix} | & | & & | \\ \vec{v}_1 - \lambda_1\vec{v}_i & \vec{v}_2 - \lambda_2\vec{v}_i & \cdots & \vec{v}_n \\ | & | & & | \end{pmatrix}$$

where the i-th column is untouched.

Dependent Columns $\Rightarrow$ det $= 0$

If one column is a linear combination of the others:

$$\vec{v}_k = \lambda_1 \vec{v}_1 + \cdots + \lambda_{k-1}\vec{v}_{k-1} + \lambda_{k+1}\vec{v}_{k+1} + \cdots$$

Subtract those multiples from column k:

$$\det(\ldots, \vec{v}_k, \ldots) = \det(\ldots,\ \vec{v}_k - \sum_{j \neq k} \lambda_j \vec{v}_j,\ \ldots) = \det(\ldots,\ \vec{0},\ \ldots)$$

But $\det(\ldots, \vec{0}, \ldots) = 0$ (scale by k=0 in P1).

Proposition

If the columns of A are **linearly dependent**, then $\det(A) = 0$.

Swapping Two Columns Flips the Sign

We already saw this for 2×2. In general:

Proposition

Swapping two columns multiplies det by -1.

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Proposition

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Proof sketch. Let $\vec{u}, \vec{v}$ be two columns. Consider:

$$0 = \det(\ldots,\ \vec{u} + \vec{v},\ \ldots,\ \vec{u} + \vec{v},\ \ldots)$$

Expand by P1 in both positions:

$$= \det(\ldots, \vec{u}, \ldots, \vec{u}, \ldots) + \det(\ldots, \vec{u}, \ldots, \vec{v}, \ldots)$$

$$+ \det(\ldots, \vec{v}, \ldots, \vec{u}, \ldots) + \det(\ldots, \vec{v}, \ldots, \vec{v}, \ldots)$$

First and last are 0 (P2). So
$\det(\ldots, \vec{u}, \ldots, \vec{v}, \ldots) = -\det(\ldots, \vec{v}, \ldots, \vec{u}, \ldots)$.

Switching Matrix

Definition

A **switching matrix** is a matrix where each row and each column has **exactly one** nonzero entry, and that entry equals 1.

Examples:

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \qquad \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \qquad \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix}$$

Its det is either $+1$ or -1. But which one?

The Zigzag Method

To find det of a switching matrix:

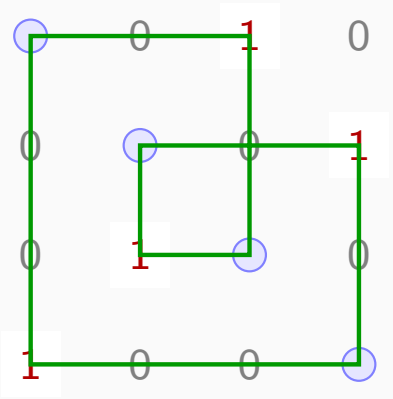
1. Draw horizontal and vertical line segments connecting each 1 to the diagonal entry in its row/column.
2. Count the number m of **connected loops**.
3. $\det = (-1)^{n+m}$

Why? Each swap breaks one loop into two. Starting from I_n (n loops, $\det = 1$), each swap flips the sign.

n loops $\to \det = (-1)^{n+n} = 1 = \det(I_n)$ ✓

Zigzag Example 1

$$S_1 = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix}$$

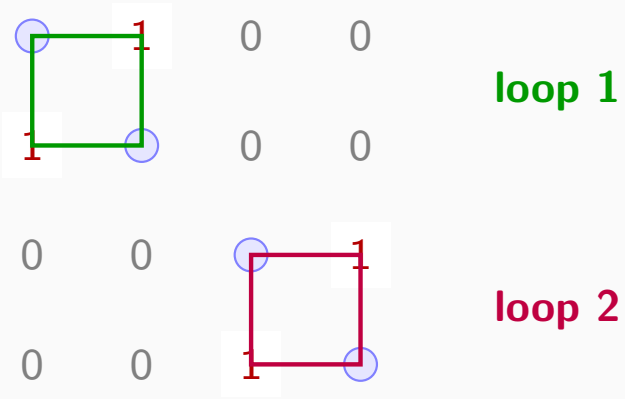


1 loop

$m = 1$ loop, $n = 4$: $\det(S_1) = (-1)^{4+1} =$ **-1**

Zigzag Example 2

$$S_2 = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$



$m = 2$ loops, $n = 4$: $\det(S_2) = (-1)^{4+2} = \mathbf{+1}$

Zigzag Practice

Find the determinant of each switching matrix:

1. $\begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}$
2. $\begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$
3. $\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \end{pmatrix}$
4. I_n always has det $=$?

Cross-Filling Computes Determinant

Cross-Filling = Column Operations

Recall: cross-filling decomposes $A = R_1 + A_1$ where R_1 is rank-1 with cross-center a_1 and A_1 has a **zero cross**.

Key insight: this decomposition is just **column operations**!

- Subtracting multiples of the pivot column from other columns zeros out the pivot row
- Column operations **preserve** det
- Then factor out a_1 from the pivot column (P1)

But A_1 has a zero at the pivot position — we cannot cross-fill A_1 directly.

The Annotated Remainder

The remainder A_1 has a **zero cross** — its pivot position is 0.

To continue cross-filling, we **annotate** that position with a 1:

Definition

The **annotated remainder** $\widehat{A}_1$ is the remainder A_1 with the pivot position replaced by 1.

The annotation **remembers** where the previous pivot was — it marks a "1 has been extracted here."

Theorem

$$\det(A) = a_1 \cdot \det(\widehat{A}_1)$$

Why? Column operations preserve det; factoring a_1 from the pivot column gives $\det = a_1 \cdot \det(\widehat{A}_1)$.

Iterate: Cross-Fill the Annotated Remainder

Now cross-fill $\widehat{A}_1$ (it is a full matrix — ready for the next pivot):

$$\widehat{A}_1 \xrightarrow{\text{cross-fill}} \widehat{A}_2 \xrightarrow{\text{cross-fill}} \cdots \xrightarrow{\text{cross-fill}} \widehat{A}_n$$

Each step extracts one cross-center a_i.

After n steps, every row and column has been used as a pivot exactly once — the final $\widehat{A}_n$ has 1's at all n pivot positions and 0's elsewhere.

That is exactly a **switching matrix** $S = \widehat{A}_n$.

Theorem

$$\boxed{\det(A) = a_1 \cdot a_2 \cdots a_n \cdot \det(S)}$$

Worked Example: Setup

$$A = \begin{pmatrix} 2 & 1 & 2 \\ 1 & 3 & 1 \\ 2 & 1 & 3 \end{pmatrix}$$

We compute $\det(A)$ by cross-filling: three steps, three colors.

Step 1: pivot at $(2, 1)$ $\rightarrow$ Step 2: pivot at $(3, 2)$ $\rightarrow$ Step 3: pivot at $(1, 3)$

Step 1: Pivot at $(2,1)$, Cross-Center $a_1 = 1$

Cross at row 2, column 1. Rank-1 piece $= \frac{1}{a_1}\text{col}_1 \times \text{row}_2$:

$$\begin{pmatrix} 2 & 1 & 2 \\ 1 & 3 & 1 \\ 2 & 1 & 3 \end{pmatrix} = \underbrace{\begin{pmatrix} 2 & 6 & 2 \\ 1 & 3 & 1 \\ 2 & 6 & 2 \end{pmatrix}}_{R_1 \text{ (rank 1)}} + \underbrace{\begin{pmatrix} 0 & -5 & 0 \\ 0 & 0 & 0 \\ 0 & -5 & 1 \end{pmatrix}}_{A_1 \text{ (zero cross)}}$$

Annotate: put 1 at pivot position $(2,1)$:

$$\det(A) = 1 \cdot \det \underbrace{\begin{pmatrix} 0 & -5 & 0 \\ 1 & 0 & 0 \\ 0 & -5 & 1 \end{pmatrix}}_{\widehat{A}_1 \text{ (annotated remainder)}}$$

Step 2: Pivot at $(3,2)$, Cross-Center $a_2 = -5$

Cross-fill the annotated remainder $\widehat{A}_1$:

$$\begin{pmatrix} 0 & -5 & 0 \\ 1 & 0 & 0 \\ 0 & -5 & 1 \end{pmatrix} = \underbrace{\begin{pmatrix} 0 & -5 & 1 \\ 0 & 0 & 0 \\ 0 & -5 & 1 \end{pmatrix}}_{R_2} + \underbrace{\begin{pmatrix} 0 & 0 & -1 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}}_{A_2}$$

$$\det(A) = 1 \cdot (-5) \cdot \det \underbrace{\begin{pmatrix} 0 & 0 & -1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}}_{\widehat{A}_2 \text{ (annotated)}}$$

Step 3: Pivot at $(1,3)$, Cross-Center $a_3 = -1$

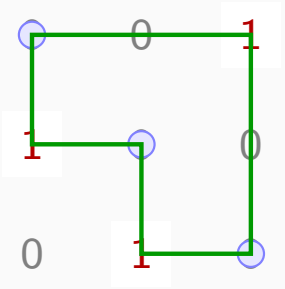
Cross-fill $\widehat{A}_2$: pivot at $(1,3)$, cross-center $a_3 = -1$.

$$\widehat{A}_2 = \begin{pmatrix} 0 & 0 & -1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix} = \underbrace{\begin{pmatrix} 0 & 0 & -1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}}_{R_3} + \underbrace{\begin{pmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}}_{A_3}$$

Annotate (put 1 at pivot): $\widehat{A}_3 = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix} = S$ — all positions used, this is the **switching matrix**!

The Switching Matrix

$$S = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}$$



1 loop

Cycle $1 \to 3 \to 2 \to 1$: one loop.

$\det(S) = (-1)^{3+1} = \mathbf{+1}$

Putting It All Together

$$\det \begin{pmatrix} 2 & 1 & 2 \\ 1 & 3 & 1 \\ 2 & 1 & 3 \end{pmatrix}$$

$$= \underbrace{1}_{\text{step 1}} \times \underbrace{(-5)}_{\text{step 2}} \times \underbrace{(-1)}_{\text{step 3}} \times \underbrace{(+1)}_{\text{switching}} = \mathbf{5}$$

Putting It All Together

$$\det \begin{pmatrix} 2 & 1 & 2 \\ 1 & 3 & 1 \\ 2 & 1 & 3 \end{pmatrix}$$

$$= \underbrace{1}_{\text{step 1}} \times \underbrace{(-5)}_{\text{step 2}} \times \underbrace{(-1)}_{\text{step 3}} \times \underbrace{(+1)}_{\text{switching}} = \mathbf{5}$$

Verify: expand along row 1:

$$2(9-1) - 1(3-2) + 2(1-6) = 16 - 1 - 10 = 5 \checkmark$$

In Practice: Just Cross-Fill

The annotated remainder $\widehat{A}$ explains **why** the formula works.

But you **don't need to write it down**. In practice:

1. Cross-fill as usual (Lecture 8) — just record each **cross-center** a_i
2. The pivot positions $(r_1, c_1), (r_2, c_2), \ldots, (r_n, c_n)$ define the switching matrix S
3. $\det(A) = a_1 \cdot a_2 \cdots a_n \cdot \det(S)$

When does it fail?

If at some step **no nonzero entry** remains for a pivot $\Rightarrow \det = 0$.

Example: Direct Computation

$$A = \begin{pmatrix} 2 & 1 & 3 \\ 1 & 3 & 1 \\ 2 & 1 & 5 \end{pmatrix}$$

Cross-fill (just record pivots):

Step 1:	pivot at $(2,1)$	cross-center $a_1 = 1$
Step 2:	pivot at $(1,2)$	cross-center $a_2 = -5$
Step 3:	pivot at $(3,3)$	cross-center $a_3 = 2$

Pivot positions: $(2,1),(1,2),(3,3) \Rightarrow S = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$, $\det(S) = -1$

$$\det(A) = 1 \times (-5) \times 2 \times (-1) = \mathbf{10}$$

Exercise

Compute by cross-filling:

$$\det \begin{pmatrix} 2 & 3 & 1 \\ 1 & 1 & 2 \\ 3 & 2 & 1 \end{pmatrix}$$

1. Cross-fill with first pivot at $(2, 1)$. Record each cross-center.
2. Read off the switching matrix from the pivot positions.
3. Use zigzag for $\det(S)$.
4. Multiply: $\det = a_1 \cdot a_2 \cdot a_3 \cdot \det(S)$.

Solution: Step 1

$$A = \begin{pmatrix} 2 & 3 & 1 \\ 1 & 1 & 2 \\ 3 & 2 & 1 \end{pmatrix}$$

Pivot at $(2, 1)$, cross-center $a_1 = 1$.

Rank-1 piece: $\frac{1}{1} \begin{pmatrix} 2 \\ 1 \\ 3 \end{pmatrix} \begin{pmatrix} 1 & 1 & 2 \end{pmatrix} = \begin{pmatrix} 2 & 2 & 4 \\ 1 & 1 & 2 \\ 3 & 3 & 6 \end{pmatrix}$

After subtraction: $\begin{pmatrix} 0 & 1 & -3 \\ 0 & 0 & 0 \\ 0 & -1 & -5 \end{pmatrix}$ (row 2, col 1 zeroed out)

Solution: Step 2

$$\begin{pmatrix} 0 & 1 & -3 \\ 0 & 0 & 0 \\ 0 & -1 & -5 \end{pmatrix}$$

Pivot at $(3, 2)$, cross-center $a_2 = -1$.

Rank-1 piece: $\frac{1}{-1}\begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}\begin{pmatrix} 0 & -1 & -5 \end{pmatrix} = \begin{pmatrix} 0 & 1 & 5 \\ 0 & 0 & 0 \\ 0 & -1 & -5 \end{pmatrix}$

After subtraction: $\begin{pmatrix} 0 & 0 & -8 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$ (only entry $(1, 3)$ remains)

Solution: Step 3 and Result

$$\begin{pmatrix} 0 & 0 & -8 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

Pivot at $(1,3)$, cross-center $a_3 = -8$. Done — all entries extracted.

Pivot positions: $(2,1)$, $(3,2)$, $(1,3)$ $\Rightarrow$ switching matrix:

$$S = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}$$

Zigzag: cycle $1 \to 3 \to 2 \to 1$. One loop. $\det(S) = (-1)^{3+1} = +1$.

$$\det(A) = 1 \times (-1) \times (-8) \times (+1) = \mathbf{8}$$

Exercise: 4×4

Compute by cross-filling:

$$\det \begin{pmatrix} 1 & 1 & 0 & 1 \\ 1 & 2 & 1 & 0 \\ 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 2 \end{pmatrix}$$

Properties of Determinant

$\det(A^\top) = \det(A)$

Theorem

For any square matrix A: $\det(A^\top) = \det(A)$.

This means: everything we said about **columns** also holds for **rows**.

- Row operations preserve det (add multiple of one row to another)
- Two equal rows $\Rightarrow \det = 0$
- Swapping rows flips sign
- Dependent rows $\Rightarrow \det = 0$

Proof via Cross-Filling

If $A = R_1 + R_2 + \cdots + R_n$ is a cross-filling for A,

then $A^\top = R_1^\top + R_2^\top + \cdots + R_n^\top$ is a cross-filling for $A^\top$.

Proof via Cross-Filling

If $A = R_1 + R_2 + \cdots + R_n$ is a cross-filling for A,

then $A^\top = R_1^\top + R_2^\top + \cdots + R_n^\top$ is a cross-filling for $A^\top$.

Key observations:

1. Transposing a rank-1 cross piece: cross-center **stays the same** (the pivot value doesn't change, just its row/column swap)

2. The switching matrix $S^\top$ has the **same loops** as S (paths just reverse direction, loop count unchanged)

Therefore: $\det(A^\top) = \underbrace{a_1 \cdots a_n}_{\text{same}} \cdot \underbrace{\det(S^\top)}_{\det(S)} = \det(A)$. $\square$

Transpose Preserves Loops

$S^\top \neq S$, but the **loop count** is the same — transposing reverses each cycle's direction without splitting or merging loops.

Why This Matters

$\det(A^\top) = \det(A)$ gives us a **duality**:

Column property		Row property
column operations preserve det	$\leftrightarrow$	row operations preserve det
equal columns $\Rightarrow$ det $= 0$	$\leftrightarrow$	equal rows $\Rightarrow$ det $= 0$
swap columns flips sign	$\leftrightarrow$	swap rows flips sign
dependent columns $\Rightarrow 0$	$\leftrightarrow$	dependent rows $\Rightarrow 0$

From now on, we can freely use **either** row or column operations when computing determinants.

Block Diagonal Matrices

Theorem

If A is block diagonal with square blocks:

$$\det \begin{pmatrix} B_1 & 0 & \cdots & 0 \\ 0 & B_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & B_k \end{pmatrix} = \det(B_1) \cdot \det(B_2) \cdots \det(B_k)$$

Why? Cross-filling gives a clean proof.

Proof via Cross-Filling

When you cross-fill a block diagonal matrix:

- Pick a pivot **inside block** B_1. The cross (row and column) stays inside B_1 — the zero blocks guarantee that cross-filling doesn't touch other blocks.
- After finishing all pivots in B_1, move to B_2, then B_3, etc.
- Cross-centers: first the pivots of B_1, then B_2, ...
- Switching matrix: block diagonal too — loops stay within each block.

So $\det(A) = \underbrace{(\text{pivots of } B_1) \cdot \det(S_1)}_{\det(B_1)} \cdot \underbrace{(\text{pivots of } B_2) \cdot \det(S_2)}_{\det(B_2)} \cdots$

Example

$$\det \begin{pmatrix} 2 & 1 & 0 & 0 & 0 \\ 1 & 3 & 0 & 0 & 0 \\ 0 & 0 & 1 & 2 & 1 \\ 0 & 0 & 0 & 1 & 3 \\ 0 & 0 & 2 & 1 & 1 \end{pmatrix}$$

$$= \det \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix} \times \det \begin{pmatrix} 1 & 2 & 1 \\ 0 & 1 & 3 \\ 2 & 1 & 1 \end{pmatrix}$$

$$= 5 \times ?$$

(Compute the 3×3 block by cross-filling!)

Block Triangular Matrices

The same argument works for block **upper** or **lower** triangular:

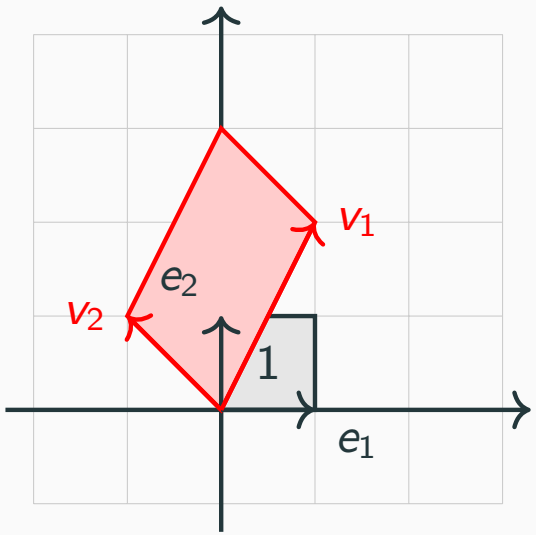
Theorem

$$\det \begin{pmatrix} B_1 & * & \cdots & * \\ 0 & B_2 & \cdots & * \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & B_k \end{pmatrix} = \det(B_1) \cdot \det(B_2) \cdots \det(B_k)$$

Why? Cross-filling block B_1 automatically zeros out the $*$ entries above it (they are in the same rows as B_1's pivots). The off-diagonal blocks get cleared for free.

In particular: det of a triangular matrix = product of diagonal entries.

Geometric Picture: Linear Maps Scale Area



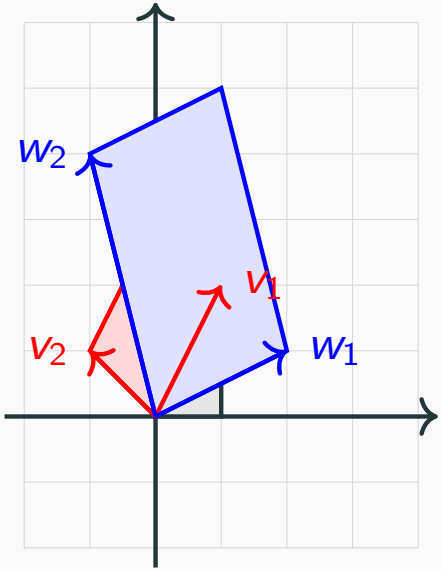
$$A = \begin{pmatrix} 1 & -1 \\ 2 & 1 \end{pmatrix}, \quad \det(A) = 3$$

A maps $e_1 \to v_1$, $e_2 \to v_2$

$$\frac{\text{red parallelogram}}{\text{unit square}} = |\det(A)| = 3$$

Determinant = **area scaling factor**

Two Transformations: Area Ratios Multiply



$$B = \begin{pmatrix} 1 & 1 \\ -1 & 2 \end{pmatrix}, \quad \det(B) = 3$$

B acts **on the A-grid**: maps $v_i \to w_i$

$$\frac{\text{blue}}{\text{red}} = |\det(B)| = 3$$

$$\frac{\text{blue}}{\text{unit}} = 3 \times 3 = 9 = |\det(AB)|$$

Area ratios **multiply**

$\implies \det(AB) = \det(A)\det(B)$

Determinant of a Product

Theorem

$$\det(AB) = \det(A) \cdot \det(B)$$

Geometric intuition: two successive linear maps scale area by $|\det A|$ then $|\det B|$.

Total scaling $= |\det A| \cdot |\det B| = |\det(AB)|$.

Now we prove it using block lower triangular + cross-filling.

Cross-Filling Reveals the Product

Recall Lecture 10: Cross-filling $\left(\begin{smallmatrix} A & I \\ I & * \end{smallmatrix}\right)$ with pivots in $A \to *$ is **passive** $\to$ reveals $-A^{-1}$.

Same idea: form the $2n \times 2n$ block matrix

$$M = \begin{pmatrix} A & 0 \\ I & B \end{pmatrix}$$

Cross-fill with pivots from I (bottom-left). The top-right block ($= 0$) is **passive** — it is never read, only written to.

Each rank-1 piece from pivot $(n{+}i,\ i) = 1$:

pivot column: $\left(\begin{smallmatrix} a_i \\ e_i \end{smallmatrix}\right)$, pivot row: $(e_i^T,\ b_i^T)$

$\to$ top-right receives $a_i \cdot b_i^T$ (one **slice of** AB)

After n steps: sum $= \left(\begin{smallmatrix} A & AB \\ I & B \end{smallmatrix}\right)$, remainder $= \left(\begin{smallmatrix} 0 & -AB \\ 0 & 0 \end{smallmatrix}\right)$

Example: Cross-Fill / from M

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}, \quad B = \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}, \quad \det(A) = 3,\ \det(B) = 5$$

$$\underbrace{\begin{pmatrix} 2 & 1 & 0 & 0 \\ 1 & 2 & 0 & 0 \\ 1 & 0 & 2 & 1 \\ 0 & 1 & 1 & 3 \end{pmatrix}}_{M=\left(\begin{smallmatrix} A & 0 \\ I & B \end{smallmatrix}\right)} = \underbrace{\begin{pmatrix} 2 & 1 & 5 & 5 \\ 1 & 2 & 4 & 7 \\ 1 & 0 & 2 & 1 \\ 0 & 1 & 1 & 3 \end{pmatrix}}_{\text{sum of 2 rank-1 pieces}} + \underbrace{\begin{pmatrix} 0 & 0 & -5 & -5 \\ 0 & 0 & -4 & -7 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}}_{\text{remainder}}$$

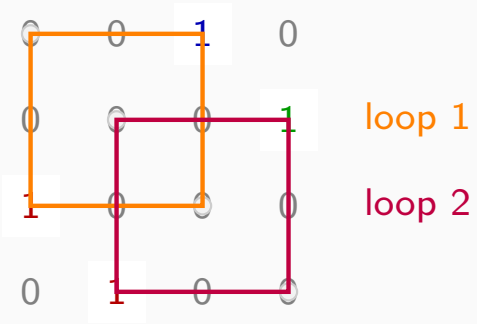
Top-right of sum $= AB = \left(\begin{smallmatrix} 5 & 5 \\ 4 & 7 \end{smallmatrix}\right)$. AB **appeared!** (Cross-centers: both $= 1$)

Example: Complete Cross-Filling

Annotated remainder: $\widehat{A}_2 = \begin{pmatrix} 0 & 0 & -5 & -5 \\ 0 & 0 & -4 & -7 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}$. Continue cross-filling $-AB$:

Step 3: pivot $(1,3)$, cross-center $= -5$. Step 4: pivot $(2,4)$, cross-center $= -3$.

Switching matrix S:



$$\det(M) = \underbrace{1 \cdot 1}_{\text{from } I} \times \underbrace{(-5) \cdot (-3)}_{\text{from } -AB} \times \underbrace{(-1)^{4+2}}_{\text{2 loops}} = 15$$

$$= \det(A)\det(B) = 3 \times 5 = 15 = \det(AB) \checkmark$$

The Two Switching Matrices

Cross-filling $M = \left(\begin{smallmatrix} A & 0 \\ I & B \end{smallmatrix}\right)$ uses pivots from two blocks:

- I-block: pivots at $(n{+}i,\ i)$ for $i = 1, \ldots, n$ (cross-centers all $= 1$)
- $-AB$ block: pivots at $(\sigma(j),\ n{+}j)$ for $j = 1, \ldots, n$

Definition

T_n is the $n \times n$ **switching matrix** of $-AB$: 1's at positions $(\sigma(j), j)$.

S_{2n} is the full $2n \times 2n$ switching matrix of M:

$$S_{2n} = \begin{pmatrix} 0 & T_n \\ I_n & 0 \end{pmatrix}$$

In the example: $T_2 = I_2$, $S_4 = \left(\begin{smallmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{smallmatrix}\right)$.

The Key Question

In the example: $S_{4\times 4}$ has **2 loops**, and the 2×2 switching matrix T_2 of $-AB$ also has **2 loops**.

Coincidence?

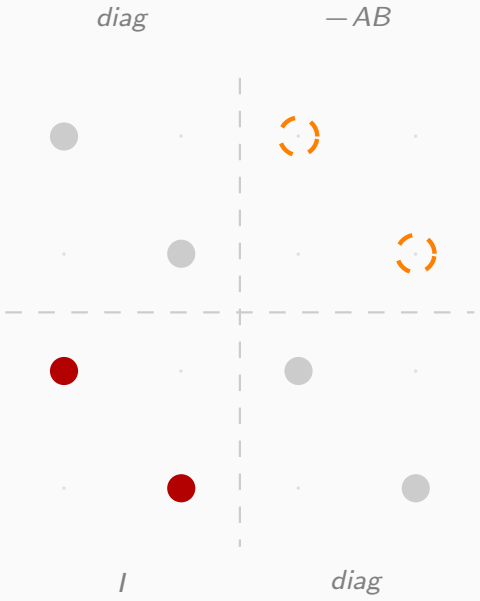
Theorem

For any σ: loops in S_{2n} = cycles of σ = loops in T_n.

Why this matters: if true, then $\det(S_{2n}) = (-1)^n \cdot \det(T_n)$, which gives us $\det(AB) = \det(A)\det(B)$.

Method: we will **deform** each loop of S_{2n} into a loop of T_n by a topological trick — replacing rectangular detours with shortcuts.

Three Diagonals



The switching matrix has three diagonals:

main : (k, k) — spans two blocks

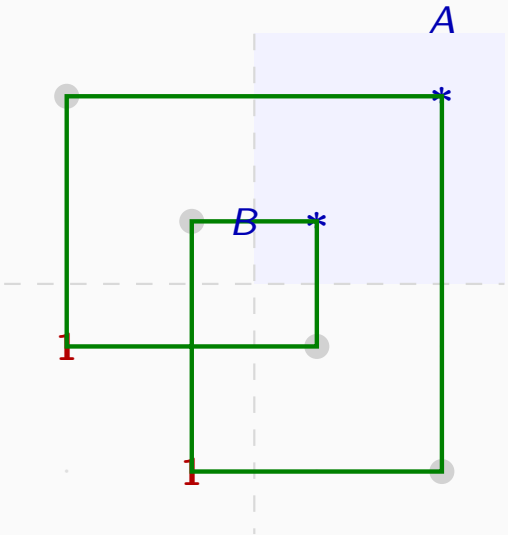
lower : $(n+i,\ i)$ — I-entries

upper : $(i,\ n+i)$ — **not yet present**

The upper diagonal will **appear** after a topological deformation.

The Rectangular Detour ($\sigma = (1\ 2),\ n = 2$)

Trace the zigzag. **Between** visits to the upper-right block, the path goes **down–left–up**:



The loop visits the upper-right block at $A{=}(1,4)$ and $B{=}(2,3)$.

From A to B, the zigzag **detours** outside:

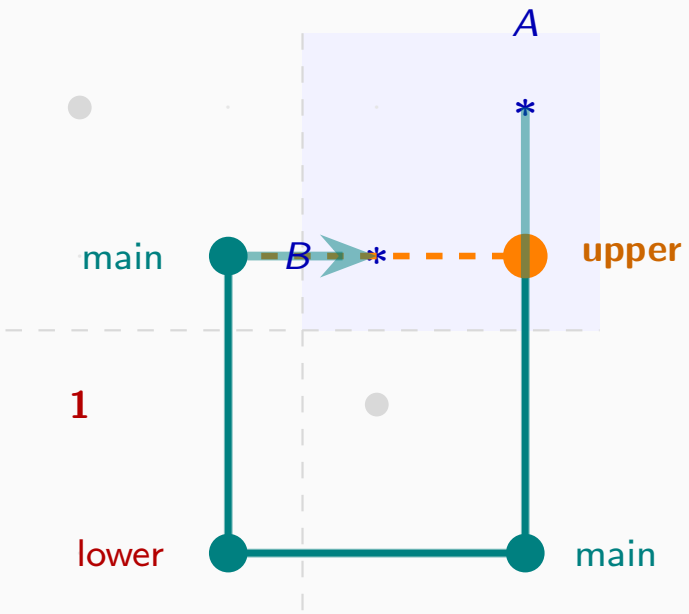
$A \xrightarrow{\text{down}} (4,4) \xrightarrow{\text{left}} (4,2) \xrightarrow{\text{up}} (2,2) \to B$

Three **corners**:

- $(4,4)$: main diagonal
- $(4,2)$: lower diagonal
- $(2,2)$: main diagonal

The Complete Rectangle

Draw the full rectangle through the three corners. The 4th corner at $(2, 4)$ sits on the **upper diagonal**:

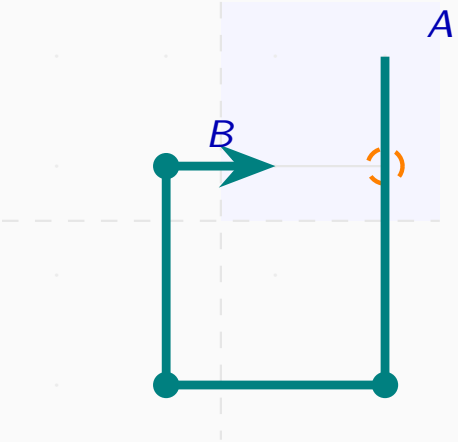


Existing (3 sides): $A \to (4, 4) \to (4, 2) \to (2, 2) \to B$ — outside the block. **Missing** (dashed): $(2, 2) \to (4, 2)$ — **through the upper diagonal**.

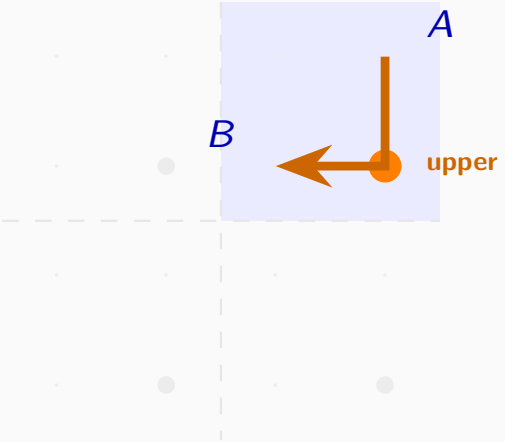
Three Corners → One Corner

Replace the 3-side detour with the missing side:

Before (3 corners, outside)



After (1 corner, inside)



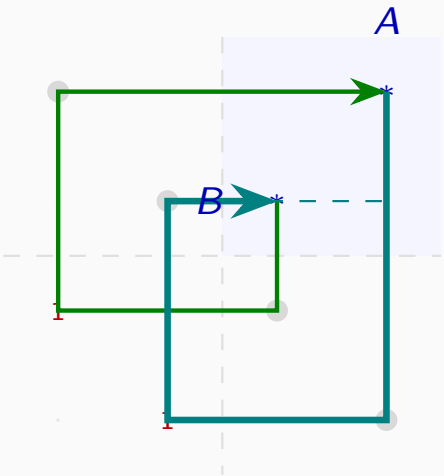
Same two endpoints A and B. The loop goes $A \to \cdots \to B \to \cdots \to A$.

We only changed one path (A to B) — the loop **does not break**.

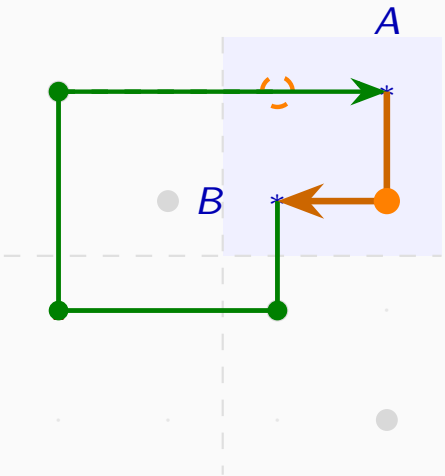
Step 1: Replace the First Detour

Deform the first rectangle ($A \to B$):

Original zigzag



After step 1

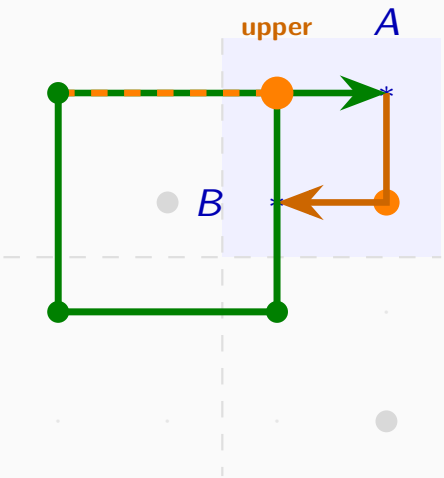


Detour 1 replaced by shortcut. The second rectangle now visible — dashed = next replacement.

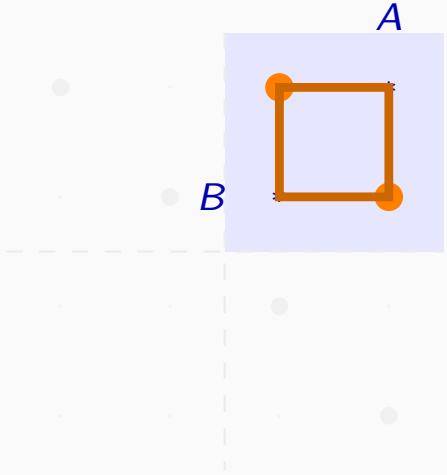
Step 2: Replace the Second Detour

Now deform the second rectangle ($B \to A$):

After step 1



After step 2: loop in block



1 loop (block only)

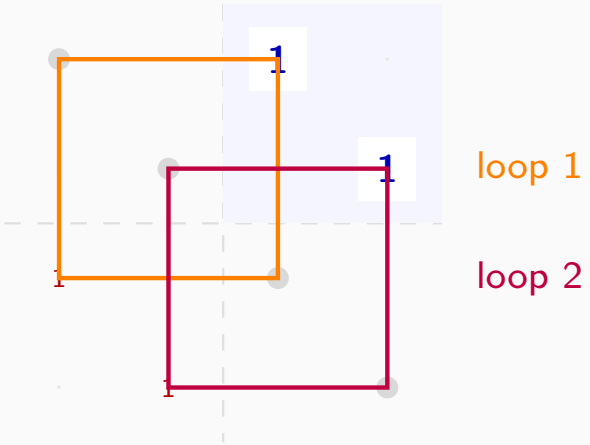
Both detours replaced. The loop now **lives entirely in the upper-right block**.

Corners alternate: $-AB$ entries and upper diagonal = a T_n **zigzag**!

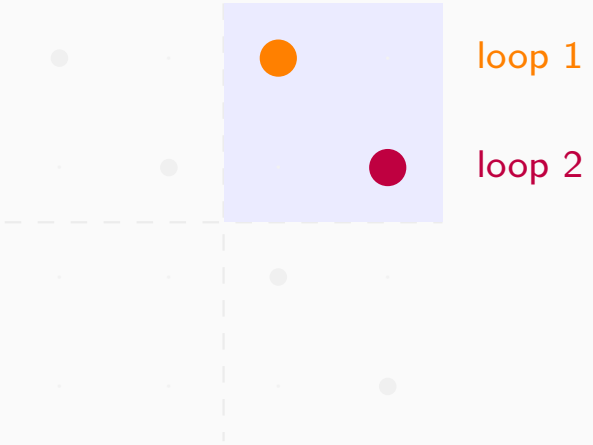
Degenerate Case ($\sigma = \mathbf{id},\ n = 2$)

$-AB$ entries at $(1,3)$ and $(2,4)$ — **already on the upper diagonal**:

Before: $S_{4\times 4}$



After: upper-right block



Each rectangle is **degenerate**: the missing 4th corner **coincides** with the $-AB$ entry itself.

Each loop collapses to a single point on the upper diagonal $\rightarrow$ **2 separate loops**.

The Reduction

	σ cycles	S_{2n} loops
n=2, $\sigma = \mathrm{id}$	2	2
n=2, $\sigma = (1\ 2)$	1	1
n=3, $\sigma = (1\ 2\ 3)$	1	1
n=3, $\sigma = (1\ 2)(3)$	2	2

Theorem

Loops in S_{2n} = cycles of σ = loops in T_n

Why? Each rectangular detour through the diagonal + I blocks can be deformed into a shortcut through the **upper diagonal**. This reduces the $2n \times 2n$ zigzag to an $n \times n$ zigzag in the upper-right block — same loops, smaller matrix.

Sign Comparison

Both S_{2n} and T_n have the same number of loops p.

$$\det(S_{2n}) = (-1)^{2n+p} \qquad \det(T_n) = (-1)^{n+p}$$

$$\frac{\det(S_{2n})}{\det(T_n)} = (-1)^{(2n+p)-(n+p)} = (-1)^n$$

$$\boxed{\det(S_{2n}) = (-1)^n \cdot \det(T_n)}$$

The deformation from $2n$ to n positions contributes exactly $(-1)^n$ to the sign — always.

Proof (General)

$$M = \begin{pmatrix} A & 0 \\ I & B \end{pmatrix}, \quad \det(M) = \det(A)\det(B) \quad \text{(block lower triangular).}$$

Cross-fill M: n pivots from I (all $= 1$), then n pivots from $-AB$.

$$\det(M) = \underbrace{1^n}_{\text{from } I} \times (\text{pivots of } -AB) \times \det(S_{2n})$$

The I-entries form n linked pairs $\to$ same loops as T_n:
$\det(S_{2n}) = (-1)^n \cdot \det(T_n)$

$$\det(M) = \underbrace{(-1)^n \cdot \det(-AB)}_{\text{pivots} \times \det(S)} = (-1)^n \cdot (-1)^n \det(AB) = \det(AB)$$

$$\boxed{\det(A)\det(B) = \det(M) = \det(AB)} \qquad \square$$

Corollary: Invertibility and Determinant

Proposition

A is invertible $\iff$ $\det(A) \neq 0$

($\Rightarrow$) If A is invertible:

$$\det(A) \cdot \det(A^{-1}) = \det(AA^{-1}) = \det(I_n) = 1$$

so $\det(A) \neq 0$.

($\Leftarrow$) If $\det(A) \neq 0$: cross-filling gives n **nonzero** pivots $\to$ rank n $\to$ invertible.

More Consequences

1. If A is invertible:
$$\det(A^{-1}) = \frac{1}{\det(A)}$$

2. For any positive integer k:
$$\det(A^k) = \det(A)^k$$

3. Recall from §2:
$$\det(\lambda A) = \lambda^n \det(A)$$